Unit 18 - Database Design
Assignment 1 - Theory and database queries

Introduction

You are working as a trainee for a development company. Your boss wants to be sure that you understand the fundamentals of database design, and database queries, before she lets you loose on an important project. To put her mind at rest, she has asked you to complete the following tasks:

Task 1(P1) - Explain the features of a relational database

Explain the main features of relational databases by copying and completing the table shown below.

What is an entity?
An entity is basically an object which contains data of a single individual, a place or a thing. The data that is contained may depend on what purpose or the intent of the data is.
What is used to store the information about several entities of one type?
There are at least 3 different methods of storing information about several entities, the entities consist of a single flat-file table, large flat-file database and data redundancy. To store a limited amount of data, a single flat-file table is used to store this data. But a large flat-file database can store much more data, but it also is more inefficient due to the fact it takes up much more space and memory on the PC. With data redundancy, the data is duplicated across records. This can occur in flat-file tables, but it can be easily avoided in relational databases.
What is a row (also known as one record) in a table?
A row (or record/tuple) is a representation of a single, completely structured data in a table. What I mean by this is that a row is one of three things that build up a table, the other two being columns and fields. Sets of related data is represented on each row in a table and the rows are all in the same structure.
What is an attribute (known as a column or field in the database)?
An attribute (or column/field) consists of a particular simple set of data that covers one for each row of the table. What I mean by this is that an attribute gives the structure according to which the data on the rows are made. Just when an attribute allows data values of a single type, doesn't necessarily mean that it only contains simple text values.
What is the datatype of an attribute? [Give an example.]
In databases, an attribute has a data type that determines what kind of value it contains. In some databases, attributes are usually floats or booleans, with vector, integer or string attributes being less common. Vector array data types are useful for animating position, velocity, acceleration, colour and other particle attributes made of three different components. For example, the attributes in an invoice might be price, number, date or paid/unpaid, etc.

Explain two advantages of getting the data type of an attribute right.

One advantage of getting the data type of an attribute right is so the database has a less chance of having an error in the database itself, which would be very time consuming if an error does occur. This could be even more time consuming because the database wouldn't be neatly built because there isn't a correct data type of an attribute.

Another advantage of getting the data type of an attribute right is that the database would be much neater than if it didn't have a correct data type. So if there is an error or a mistake within the database, it would be much less time consuming to find the error/mistake itself. Having the database neater would also reduce the amount of frustration built when trying to make a query or using the database for other reasons.

What is a validation rule? [Give an example.]

A validation rule checks to ensure that data entered into the computer is compatible. This can be done easily and quickly due to the fact that this process is automated and the software can validate data while it is being entered into a computer. The main purpose of data validation is to spot an error within the database, this means that the validation works by checking the data and if it's in accordance with a set of rules.

For example, if you have a database on 'Healthy Living' clubs, containing information of the organisation, if it is local, town, postcode, telephone number, web address and the support that it offers. In most databases, you will have to select the 'Design' button, and it will open the table so you can edit the details. You select the field to which you want to add a validation rule, eg 'Support offered' and select the 'General' menu page. The 'Support offered' can be NA (Nutritional advice), EP (exercise plan), MA (motivational advice) or HP (health plan). The validation rule is set so that only those specific options can be entered.

Why are validation rules important?

Validation rules are important because it can be used to spot an error in the database, which is its main purpose. So, if there was an error in the database, it would be very time consuming trying to find that error manually. If you had limited time to finish a database and the error cannot be found, then the database would be useless due to the fact it is not working properly. With the validation rules, the error in the database could be found in a significantly faster way.

What is a primary key?

In a database, the primary key is a field in a table which uniquely identifies each row/record in a table. Primary keys must contain unique values. A primary key column can't have null values. A database table can only have one primary key, which could either contain single or multiple fields. If a primary key is used with multiple fields, it is called a composite key.

What is a foreign key?

In a database, the foreign key is a key that is used to link two different tables together. What I mean by this is that it is a column or a combination of columns whose values match a Primary key in a different table. For example, if there are 2 tables, then the relationship between the tables matches the Primary key in one of them and a Foreign Key in the second one.

What is a “One to Many” relationship?

A “One to Many” relationship basically is when one entity is relating to many different entities. What I mean by this, is that the one entity in a table is linked with many other entities that is in another table, so that they could be used to connect information on tables in a database. For example, ‘one’ cinema has ‘many’ customers.

What is a “One to One” relationship?

A “One to One” relationship is when one entity is only relating to another entity. What I mean by this, is that the one entity in a table is linked with one another entity in a separate table. Like other relationships, this is used to connect the tables in the database. For example, ‘one’ person has ‘one’ address.

What is a “Many to Many” relationship? [Note: this cannot be done in a relational database.]

A “Many to Many” relationship is when there are many entities which are relating to many other entities. What I mean by this, is that the many entities in a table are linked with many other entities on a separate table. For example, ‘many’ subjects can be taken by ‘many’ students.

What is normalisation?

Normalisation is when you analyse on how you can make the databases much more efficient by using separate tables to reduce the redundant data. If a database has been normalised, you will notice that the data is broken down into much smaller tables and relationships are used to link them.

What is an entity relationship diagram?

In databases, an entity relationship diagram or an ER diagram, is basically a graphical representation of entities and the relationships that the entities have to each other. An ER diagram is used to contain data of the organization within the databases or other information systems.

Why does using a relational database mean less duplication (less redundant data) than using a single table (flat file) database?

The reason that a relational database means less duplication than using a single table database is due to the fact that the data within a relational database will become related and, because of this, all the data within the relational database would be relevant with each other, unlike the single table database which does not have this.

Why does a relational database with data entry using forms lead to better quality data than storing information in a paper diary or Word document?

The reason for this is due to the fact that relational database is a much more in depth software for data than word or a paper diary. Comparing relational database to paper diary or a word document for producing high quality data, is like comparing adobe photoshop to ms paint for a high quality digital image software.

Why does use of a relational database allow easier searching and sorting of data than use of a paper diary or Word document?

Since databases was specifically made to contain and access data. It would make sense if a relational database was easier to access data than a paper diary or a word document. The reason for this is due to the fact that in database you can search for a specific table, and in the table you can search for a specific entity. In a word document or paper diary, you would have to manually scroll through the pages to find a specific entity.

Task 2 (M1) - Explain referential integrity and the purpose of primary keys in building the relationships between tables

Explain "referential integrity and the purpose of primary keys in building the relationships between tables" by answering the following questions using as an example a car sales database which has a customers table (primary key CustomerID) related to a cars table (primary key Registration Number):

What information would you store in the cars table as a foreign key in order to relate each car to a single customer and why is this needed?

Since this is a store that you can purchase cars from and each car can only be sold to a single customer because only one customer can own one car. So, the information that I will have to store is: the manufacturer of the car, the model of the car, the price of the car, colour of the car, etc for the cars table. For the customer's table I will need to have information such as: customer first name, customer last name, email address, phone number, etc.

The reason this is needed to so that there is enough information and data to know exactly which customer is going to get which car. This type of data can also be in a referential integrity relationship which would make it a lot easier to gather data, I will be explaining referential integrity later on.

Why is it important that the primary key in the customers table is unique?

The reason that the primary key in the customers table is unique is that the data does not get duplicated in two rows in the database. What I mean by this is that one row in the database is allowed to have null for the value of the unique key constraint. Although a table should have a PK, it doesn't need to have any additional unique keys.

This is important due to the fact that customers almost always don't have the same name, let alone the same car. So removing the ability of duplication in the customers table would make it less frustrating and confusing, but a worse error could be having the wrong information. Having the wrong information is without a doubt a lot worse than having no information at all.

What does it mean if referential integrity is enforced (the "Enforce referential integrity" tickbox is ticked) in the relationship between customers and cars?

When the “Enforce referential integrity” tickbox is ticked, it means that it has a referential integrity relationship which basically means that each value that has been entered in a foreign key column matches an existing value in the related primary key column. What I mean by this is that the Referential integrity is a property of data which requires every value of one attribute of a table to exist as a value of another attribute in either a different or similar table.

This means that the referential integrity relationship between the customers and the cars are much neater compared to if it didn't have a referential integrity. With referential integrity, every customer has his own car. By this I mean that if there is a car without a customer then it wouldn’t appear in the referential integrity relationship, which would make the data on a table easy to gather information on without too much clutter from useless or unneeded information.

What could go wrong as a result of not enforcing referential integrity (the “Enforce referential integrity” tickbox is not ticked) in the relationship between customers and cars?

If the “Enforce referential integrity” tickbox is not ticked, it means that it will loose its enforced referential integrity, which would make it a lot less neater that if it was on. This would cause many problems such as having a lot of useless data which would clutter the database and take up storage and memory space on the PC, which could impact the performance of the PC.

It would also make it very difficult to find information about what customer has what vehicle, which would be a big problem because it’s very time consuming and frustrating if you want to collect simple information about a car or a customer.

How does the use of the relationship between foreign key in the cars table linked to the primary key in the customers table help when creating queries and reports?

The relationship between the foreign key in the cars linked to the primary key in the customers table helps creating queries and reports due to the fact that the foreign key is a reference to the primary key, so the relationship will work well and not create errors within the database.

Since the foreign key is a good reference to the primary key, that also means that the table would be set up neatly so the information about the data could be easily read without confusion or frustration from being cluttered.

Task 3 (D1) - Discuss how potential errors in the design and construction of a database can be avoided

Written report of between 1 and 2 A4 pages that discusses:

At least two things that might go wrong in designing a database.

Firstly, the most obvious thing that will go wrong in the designing of the database is the failing or lack of planning the database. Having a careful planning phase is the only way to ensure that you didn’t end up with a database hacked together with components that failed to take into account data integrity and system performance.

If the necessary amount of time and effort isn't invested in the beginning to understand the data, the constraints and the processes, then it would be highly likely that the database would need to be refactored, replaced or even abandoned altogether.

Secondly, it's not bothering with the documentation. If a database and its data environment is not documented properly, it would be very difficult to maintain or change them when the business decides to make a change to the requirements. What I mean by this, is that I'm not only mentioning the modeling diagram, I am also mentioning data dictionaries, extended properties, data flow diagrams, details about the data sources and how data is transformed.

Having a lack of proper documentation might be very disadvantageous due to the fact that you have very little planning or very little thought of the documentation and the understanding of how significant it is, it could impact the production of the database, and it would be very likely that something might go wrong.

At least two things that might go wrong in constructing a database.

Firstly, is using poor naming conventions. This may be obvious for some, but for others it isn't as obvious or important. However, if you make a table with the names as "st_cu" or "storc", for "store customer" or column names such as "sft_nm" or "Fst", for "first name, this would be very frustrating when making queries or a schema. It is important that the name that is written down provides an introduction to the data and it should make somewhat logical sense.

Inconsistent naming conventions and poor name choices have no point whatsoever and it only causes frustration to those who need to access the database you have produced. Also, creating names that has to be limited due to the fact that they include spaces or have reserved keywords, such as "check" or "statistics" or "vertified stats". This would make it very difficult to read and frustrating.

Secondly, it's not knowing how to normalize. This is important because a properly normalized database can help improve the performance of the PC. Denormalizing too often or too soon can result in overly complex queries and locking contention as well as put the data's integrity at risk. A normalized data structure is integral to effective database design and deviating from this strategy should be done only after carefully analyzing the situation and measuring the available alternatives.

Normalisation provides the structure necessary to effectively access, modify and manage the data while ensuring its integrity over the long term.

Good practice that should be implemented to minimise all the problems discussed above
To minimise the problem of failing to plan, is to simply just plan a lot more carefully before creating your own database. Do this by rethinking your approaches to design and development of a database and invest the necessary time and effort in the beginning to understand the data, the constraints and the processes.

To minimise the problem of not bothering with the documentation, is to make the documentation a bigger priority than you initially thought. Since the documentation is important for a database,

it is important to make it a big priority. Do this by completing the data dictionary and make it accurate and kept up-to-date as the components change.

To minimise the problem of poor naming conventions, is to reduce the amount of poor names that you put on a database. You do this by coming up with a consistent naming standard that's used throughout the database. The important part is that you follow a consistent naming standard throughout and you use descriptive names so anyone who views the schema and queries can easily pick up on what's going on.

To minimise the problem of not knowing how to normalize, is to understand how normalizing works before you can design databases effectively. You do this by starting to learn the rules of normalization and their importance. If you're designing relational databases without knowing the difference between normal forms and Excel workbook, then you should make learning what normalization a priority.

Task 4 (P5 also U3P7, U3M3) - Perform queries using multiple tables and multiple criteria

Using the Prime Ministers Of Australia database system (available on Moodle), produce the following two queries.

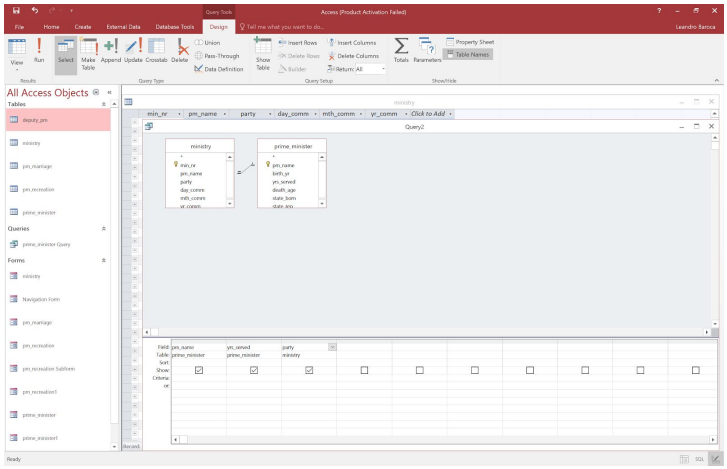
Query 1: A list of Prime Ministers who were in the Labor party (see Ministry table), and who are still alive

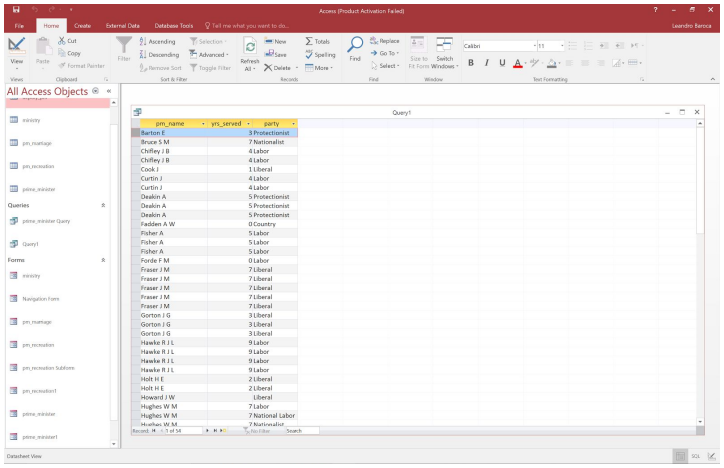
Display (only) the following fields

- *Prime Minister name*
- *Years Served*
- *Party*

Save the query with the name: ***Query1***

Provide screen prints of the query in Design View, and of the results.





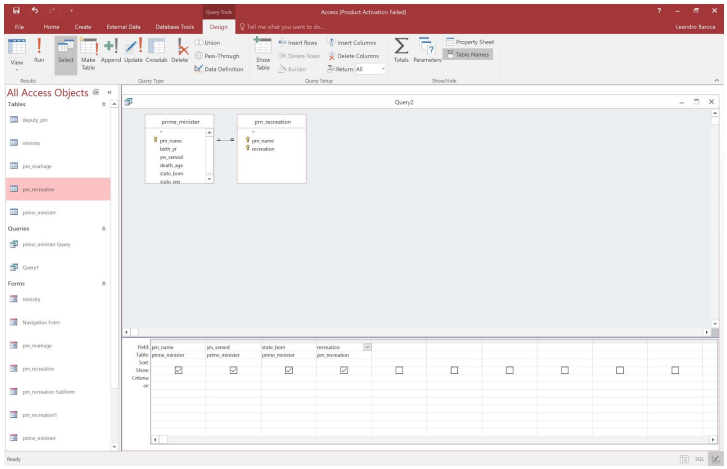
Query 2: A list of Prime Ministers who are listed as having a stated recreation that is NOT golf, who were born in either VIC or NSW, and who served more than three years in office.

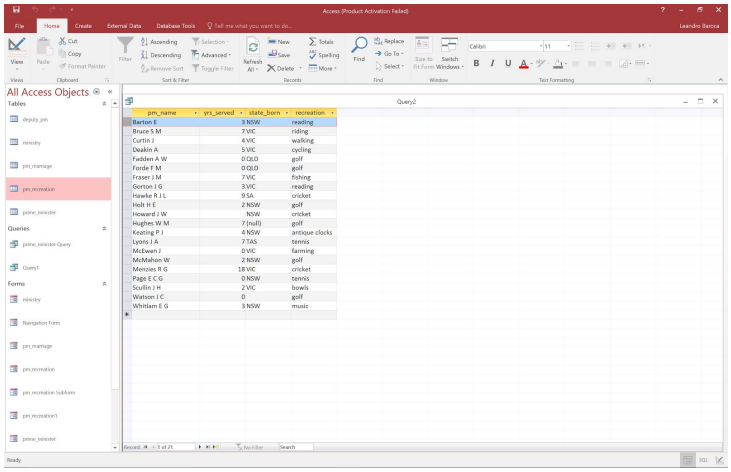
Display (only) the following fields

- *Prime Minister name*
- *Years Served*
- *State Born*
- *Recreation*

Save the query with the name: ***Query2***

Provide screen prints of the query in Design View, and of the results.



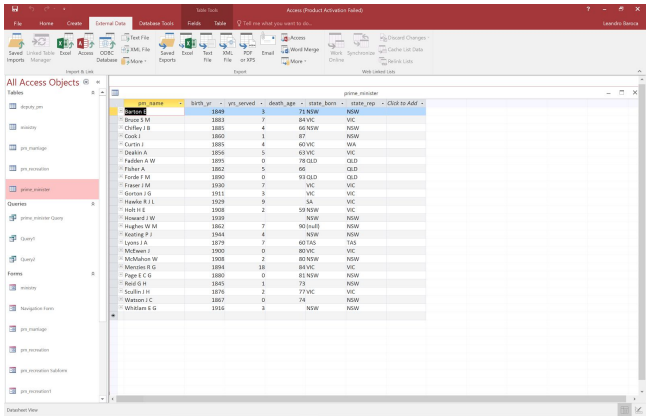


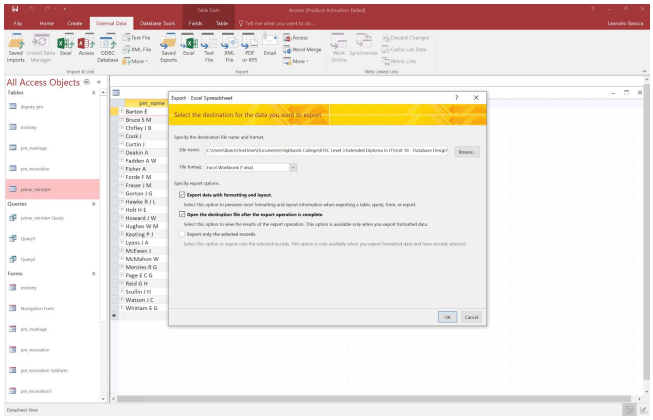
Task 2 (M2) - Import data from an external source

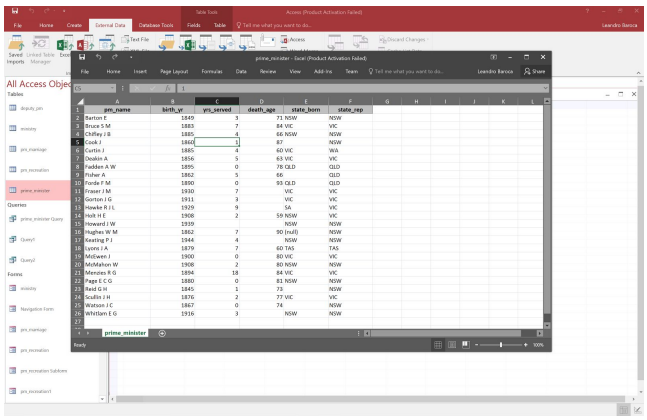
Using the Export features of MS Access, export the Prime_Minister table into an Excel spreadsheet.

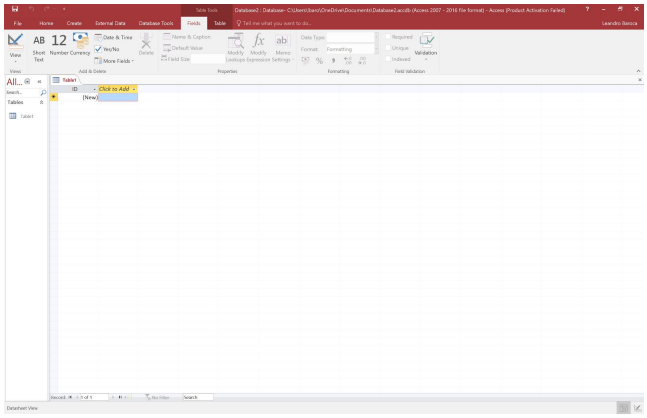
Create a new MS Access database, and import this spreadsheet into it.

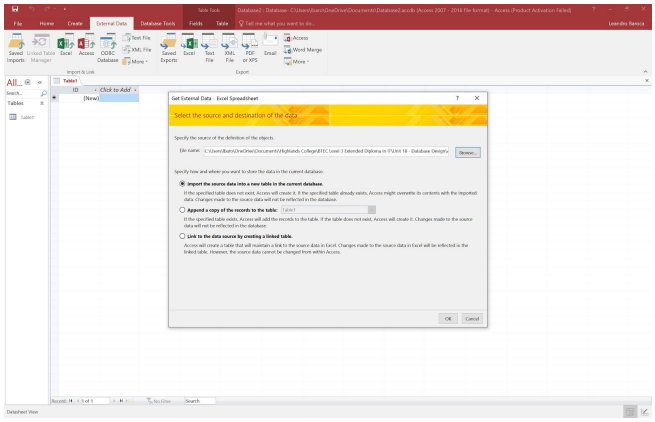
Show screen shots for each of the steps above.

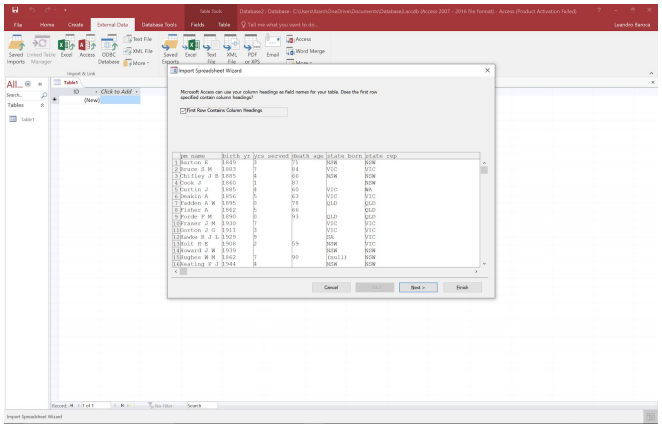


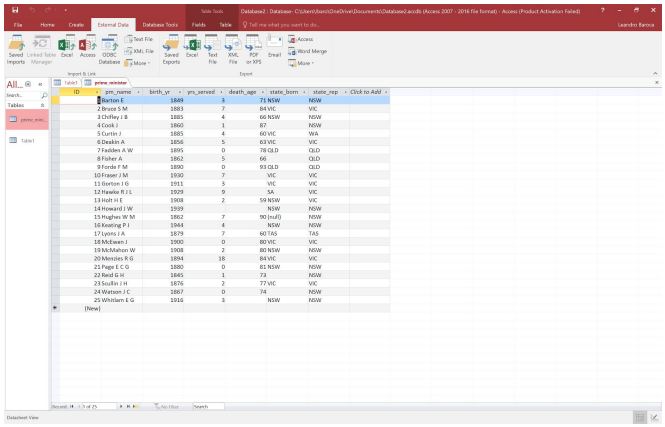












Task 3 [M3] - Export data to an external source

Produce an EXCEL file showing the Exported results for Query1 and Query2 (as per your queries above)

Show screenshots of each of these exported Excel files.

